Project Concept Idea

Student ID:	B01042273
Name:	Steven Eleojo Onoja
University email:	Onoja-se@ulster.ac.uk
Project Title:	The Application of Deep Learning in Spectral Data-Based Food Analysis: A Citizen-Deployable Framework for Food Fraud Detection

Project Summary

Food fraud, the deliberate adulteration, substitution, or mislabelling of food products for economic gain , represents a growing global challenge with serious public health, economic, and ethical implications. Conventional laboratory-based detection methods, while accurate, are costly, time-consuming, and inaccessible to most consumers and small-scale regulators. Near-infrared (NIR) and Raman spectroscopy, delivered via portable, miniaturised spectrometers, offer a compelling alternative: a low-cost, non-destructive means of capturing compositional spectral signatures from food products at scale.

This project proposes developing and evaluating a deep learning pipeline to classify food fraud events from raw spectral data collected with portable spectrometers. Convolutional neural networks (CNNs) and/or Long Short-Term Memory (LSTM) networks will be applied to publicly available spectroscopic datasets to distinguish authentic products from adulterated counterparts. The pipeline will be benchmarked against classical machine learning baselines (SVM, Random Forest) and will incorporate SHAP-based interpretability to surface the spectral regions most indicative of fraud.

A successful outcome will demonstrate that deep learning can meaningfully outperform traditional approaches on spectral food data, and that the resulting model is deployable in low-resource, citizen-science contexts, advancing the democratisation of food fraud detection.

Literature Review

The following five peer-reviewed works contextualise this research and define the gap it addresses:

[1] Mishra, P. et al. (2021). "Deep Chemometrics for the Interpretation of Spectroscopic Data." Comprehensive Reviews in Food Science and Food Safety (Wiley/IFT). Demonstrates CNN superiority over PLS-DA on NIR food data; foundational for this project's model selection rationale. DOI: 10.1111/1541-4337.12492

[2] Cama-Moncunill, X. et al. (2021). "Deep Learning-Based Approaches for Foodborne Pathogen Detection Using Hyperspectral Imaging." Sensors. Establishes precedent for CNN application to spectral food safety data; informs architecture design choices.

[3] Bala, R. et al. (2022). "Food Fraud Detection Using Near-Infrared Spectroscopy and Machine Learning." Food Control. Provides direct ML baseline benchmarks (SVM, PLS-DA) against which this project's deep learning models will be compared.

[4] Lundberg, S.M. & Lee, S.I. (2017). "A Unified Approach to Interpreting Model Predictions." NeurIPS. Foundational SHAP methodology; adopted in this project to produce interpretable spectral feature attributions for non-expert users.

[5] Qu, J.H. et al. (2023). "Portable NIR Spectrometers for Food Adulteration Detection: A Review." Food Chemistry. Reviews the current landscape of miniature spectrometer deployment for food fraud; directly motivates the citizen-science deployment framing of this project.

Research Gap: While deep learning has been applied to spectral data in controlled laboratory settings, no published study has developed an end-to-end pipeline explicitly designed for deployment via portable, consumer-grade spectrometers in a citizen-science context. The interpretability of deep learning predictions for non-expert users remains underexplored. This project addresses both gaps.

Methodology

Project Concept Idea

Phase 1 , Data Acquisition and Preprocessing:
Publicly available spectroscopic datasets will be sourced, including: the LUCAS Soil Spectral Library (NIR/VNIR), the open-access olive oil adulteration dataset (UFSC/UCI), honey adulteration NIR datasets (Kaggle/published supplementary data), and the Meat Species Authentication dataset. Data will be preprocessed using standard chemometric techniques: baseline correction (Savitzky-Golay smoothing), standard normal variate (SNV) normalisation, mean centering, and train/validation/test splitting (70/15/15 stratified by class).

Phase 2 , Feature Engineering and Exploratory Analysis:
Principal Component Analysis (PCA) will be applied for dimensionality visualisation and to inform input structuring for deep learning models. Spectral regions of interest will be identified via domain literature before model training to guide interpretability analysis.

Phase 3 , Model Development and Training:
Three model architectures will be implemented and compared: (1) Classical baselines, SVM with RBF kernel and Random Forest (200 estimators), representing established chemometric benchmarks; (2) 1D-CNN, a convolutional architecture treating spectral data as a 1D signal sequence, with hyperparameters tuned via Keras Tuner; (3) LSTM or BiLSTM, treating the spectral wavelength sequence as a temporal series. All models will be evaluated via 5-fold stratified cross-validation. TensorFlow/Keras and scikit-learn will be used as primary frameworks.

Phase 4, Interpretability Integration:
SHAP DeepExplainer and GradientExplainer will be applied to the trained CNN/LSTM models to produce wavelength-level feature importance attributions. These will be visualised as spectral contribution plots , overlaying SHAP values onto spectral curves , enabling non-expert interpretation of which spectral regions drive fraud classification decisions.

Phase 5, Lightweight Deployment Interface:
A Streamlit web application will present: fraud probability score, top contributory spectral bands, product authenticity verdict, and a batch upload facility for CSV spectral exports from portable spectrometers. The interface will be designed for use by non-specialist operators (food inspectors, citizen scientists, small retailers).

Evaluation Metrics:
Quantitative: Accuracy, Precision, Recall, F1-score, AUC-ROC, Matthews Correlation Coefficient (MCC).

Any specific details and requirements

Datasets:

- Olive oil adulteration NIR dataset, publicly available (UCI/published supplementary)
- Honey authenticity NIR dataset, Kaggle or published open-access supplementary data
- Meat species authentication spectral dataset, publicly available
- LUCAS Soil NIR Spectral Library, open access, large-scale benchmark

Python Libraries:

- TensorFlow / Keras, deep learning model development (CNN, LSTM)
- scikit-learn, SVM, Random Forest baselines, cross-validation, metrics
- SHAP, model interpretability (DeepExplainer, GradientExplainer)
- pandas, NumPy, data manipulation and preprocessing
- Matplotlib, Seaborn, Plotly , spectral visualisation
- Streamlit, deployment interface
- Keras Tuner, hyperparameter optimisation

Ethics and Data Access Risk:
Very low. All datasets are publicly available open-access resources. No human subjects are involved. No IRB/ethics approval is required. All deep learning frameworks are freely available and well-documented.

Key Background Reading:
https://ift.onlinelibrary.wiley.com/doi/10.1111/1541-4337.12492

Project Concept Idea